

Name:

BUID:

Workshop 2: Matrices of Linear Transformations

1. Suppose T is a linear transformation with the following input-output behavior: $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 9$ $T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = 2$
 - (a) Determine the domain and codomain of T .
 - (b) Determine the value of $T\left(\begin{bmatrix} 2 \\ -3 \end{bmatrix}\right)$.
2. Suppose T is a linear transformation with the following input-output behavior: $T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ $T\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$
 - (a) Determine the domain and codomain of T .
 - (b) Determine the following values: $T\left(\begin{bmatrix} 4 \\ 6 \end{bmatrix}\right)$ $T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$ $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$ $T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$
 - (c) Determine the matrix that implements T .
 - (d) Is the $\text{ran}(T) = \text{cod}(T)$? Explain your answer.
3. Determine the matrix that implements rotation by 90 degrees clockwise about the origin in $\mathbb{R}^2$.
4. Determine the matrix that implements the following transformation: $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + x_2 \\ x_1 \\ x_2 - 2x_1 \end{bmatrix}$
5. Determine the matrix that implement rotation about the z -axis in $\mathbb{R}^3$ counterclockwise 180 degrees, followed by reflection across the xy -plane.
6. Let A be the matrix $\begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$. Draw the the image of the unit square under the transformation $\mathbf{x} \mapsto A\mathbf{x}$.
7. Determine a matrix B , such that $A \neq B$ and the image of the unit square under the tranformation $\mathbf{x} \mapsto B\mathbf{x}$ is the same as the image under $\mathbf{x} \mapsto A\mathbf{x}$ (i.e., if you replace A with B in the previous problem, you would get the same drawing).

Write your solutions below and on the back on this page.

Continue your work here.